

► **Figure 25.5 A gallery of fossil types.** (a) **Sedimentary rock.**

Most fossils are found in sedimentary rocks, such as this plant fossil. (b) **Mineralized organic matter.** Some fossils, such as this petrified tree trunk, form as minerals seep into and replace organic matter. (c) **Trace fossils.** Footprints, burrows, or other traces of an organism's activities can be preserved in the fossil record. (d) **Amber.** Entire organisms are sometimes found preserved in hardened resin from a tree.

(e) **Frozen soil, ice, and acid bogs.** Rarely, frozen soil, ice, or an acid bog can preserve the body of larger organisms, such as this wolf pup.

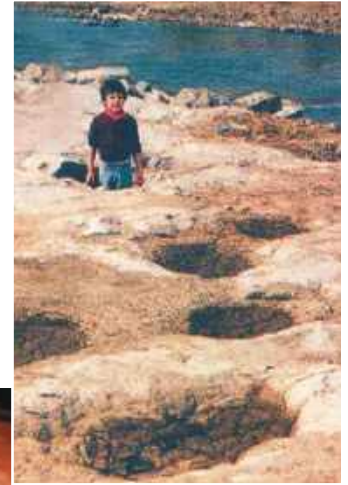
(a) Plant fossil, about 300 million years old



(b) Petrified tree in Arizona, about 215 million years old



(c) 150-million-year-old dinosaur tracks in Colorado



(e) A 50,000-year-old wolf pup found in frozen soil in Yukon, Canada



(d) Insect in amber, about 40 million years old

form, how scientists date and interpret them, and what they can and cannot tell us about changes in the history of life.

The Fossil Record

Sedimentary rocks are the richest source of fossils. As a result, the fossil record is based primarily on the sequence in which fossils have accumulated in sedimentary rock layers, called *strata* (see Figure 22.3). Useful information is also provided by other types of fossils, such as insects preserved in amber (fossilized tree sap) and mammals frozen in soil (Figure 25.5).

The fossil record shows that there have been great changes in the kinds of organisms on Earth at different points in time. Many past organisms were unlike organisms living today, and many organisms that once were common are now extinct. As we'll see later in this section, fossils also document how new groups of organisms arose from previously existing ones.

As substantial and significant as the fossil record is, keep in mind that it is an incomplete chronicle of evolution. Many of Earth's organisms did not die in the right place and time to be preserved as fossils. Of the fossils that were formed, many were destroyed by later geologic processes, and only a fraction of the others have been discovered. As a result, the known fossil record is biased in favor of species that existed for a long time,

were abundant and widespread in certain kinds of environments, and had hard shells, skeletons, or other parts that facilitated their fossilization. Even with its limitations, however, the fossil record is a remarkably detailed account of biological change over the vast scale of geologic time. Furthermore, as shown by recently unearthed fossils of whale ancestors with hind limbs (see Figure 22.19 and Figure 25.1), gaps in the fossil record continue to be filled by new discoveries.

How Rocks and Fossils Are Dated

Fossils provide valuable data for reconstructing the history of life, but only if we can determine where they fit in that story. While the order of fossils in rock strata tells us the sequence in which the fossils were laid down—their relative ages—it does not tell us their actual ages. Examining the relative positions of fossils is like peeling off layers of wallpaper in an old house. You can infer the sequence in which the layers were applied, but not the year each layer was added.

How can we determine the age of a fossil? One of the most common techniques is **radiometric dating**, which is based on the decay of radioactive isotopes (see Concept 2.2). In this process, a radioactive “parent” isotope decays to a “daughter” isotope at a characteristic rate. The rate of decay is expressed by the **half-life**, the time required for 50% of the parent